

Routing Markets as Property-Tested Multi-Agent Environments

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Abstract

AI inference routers allocate token demand among strategic providers, but their clearing state, capacity, and quality are partly hidden. This makes a simulator easy to build and difficult to validate. We introduce a request-level multi-agent environment whose providers jointly choose quotes, admission, and availability against pluggable routing mechanisms. The environment enforces exact settlement, capacity, fallback, seeded common random numbers, and provider-private feedback through a parallel RL API. We pair it with a property ladder: mechanism invariants, held-out market moments, exact simulator interventions, adversarial deviations, and live transport gates. Two studies show why the ladder matters. Preserving reward marginals while shuffling only signal order raises buyer price by 0.118 in a focal UCB/UCB game, but by -0.00047 across heterogeneous learners. A hardened adaptive router sharply reduces one-shot quote and sybil manipulation, yet leaves 9.68 times more normalized post-UCB exploitability in its first learning screen. Public quote data show nominal residual coupling, but concentration and incomplete downstream tests block transport to provider conduct. The result is an executable standard for falsifying routing mechanisms, not a claim that simulated agents reproduce live providers.

1 Introduction

AI inference is becoming a strategic allocation problem. A harness sends a request naming an open-weight model; a router chooses among providers that run that model; and providers choose price, admission, and service quality. The objects are partly substitutable but not identical. Sampling, quantization, tool-call support, latency, privacy, and hidden capacity alter the delivered product even under a common model label.

This is a natural multi-agent learning environment. Providers repeatedly act, receive private profit feedback, and adapt against a router that transforms their actions into demand. It is also a dangerous environment to study naively. A public price menu is not a firm execution commitment. A documented inverse-price score is not necessarily realized selection after eligibility and fallback. A simulator can generate clean welfare and collusion statistics even when the corresponding live quantities are unobserved.

We propose a different standard: a routing simulator should be *property-tested*. Its internal invariants must hold exactly; its observations must match the information available to real agents; its calibration targets must survive holdout; causal simulator interventions must preserve declared nuisance structure; and a separate transport gate must pass before interpreting a simulated mechanism in the live market. Failed gates are results rather than inconveniences.

We implement this standard in `orcap`, a deterministic request-level kernel with a lightweight parallel multi-agent API. A provider action contains a quote, admitted-capacity fraction, and availability state. A router plugin returns an attempted-provider order. Settlement applies capacity constraints, technical failure, fallback latency, payment, variable cost, capital cost, user utility, and welfare request by request. Stable hash-derived substreams give common random numbers across mechanism arms. Provider observations expose public quotes and only the provider’s own realized settlement; rival cost, capacity, and reliability remain private.

The environment includes random, inverse-price, quality-adjusted, price-cut memory, menu-projected, and hardened adaptive routers. Provider policies include static regimes, calibrated species, tabular Q-learning, UCB, epsilon-greedy, and an optional LLM adapter. Global finite-grid best responses, two-provider coalitions, identity splitting, fading quotes, and sequential response oracles audit strategic robustness.

Our evaluation has three main findings.

1. An exact signal-order intervention validates a narrow Hansen–Misra–Pai-style mechanism [Hansen et al., 2021]. In a focal two-UCB prisoner’s-dilemma game, preserving common signal order increases exploration coupling, joint high-price play, and buyer price. The two-seed factorial has 1,440 rows and 720 paired cells; the buyer-price effect is $+0.118$. The effect disappears under UCB/epsilon/static mixtures (-0.00047). The executable verdict is therefore negative on transport.
2. Hardening an adaptive score with lagged inputs, commitment epochs, operator-level caps, and one-sided share trust regions works mechanically. Across 4,878 historical menus it reduces mean maximum quote-attack gain from 0.226 to 0.030

and removes the measured identity-splitting gain. Yet the first UCB screen ends at profiles with normalized residual exploitability 3.929 versus 0.406 under inverse-square routing. Protecting a rule from one-step attacks can make its learning dynamics harder.

3. Public and owned data sharply limit external interpretation. A residual- coupling statistic reaches an unadjusted $p = 0.04998$, but one provider pair accounts for 87.3% of paired events, two later mechanism legs are power-gated, and the multiplicity family is incomplete. We do not identify collusion, communication, or provider algorithm.

The contribution is methodological and executable. Existing multi-agent frameworks provide general APIs [Terry et al., 2021, Lanctot et al., 2019]; algorithmic-pricing work demonstrates emergent coordination in stylized demand systems [Calvano et al., 2020]. We add request-level inference-market accounting, mechanism plugins, empirical property gates, and negative transport results.

2 The Strategic Routing Environment

2.1 Timing and primitives

There are providers $i \in \{1, \dots, n\}$. Provider i has private technology

$$\xi_i = (c_i, K_i, F_i, \rho_i, L_i),$$

where c_i is attempt cost, K_i physical capacity, F_i capital cost per slot, ρ_i reliability, and L_i base latency. At epoch t , it chooses

$$a_{it} = (p_{it}, \kappa_{it}, z_{it}),$$

where $p_{it} > 0$ is its quote, $\kappa_{it} \in [0, 1]$ its admitted-capacity fraction, and z_{it} is active, degraded, or withdrawn. Effective capacity is the integer floor of $K_i \kappa_{it}$ times a degradation multiplier.

A workload specifies input and output tokens, delivered value v , failure loss ℓ_F , latency cost ℓ_L , fallback latency, and a maximum number of attempts. A router observes public actions and whatever attributes its plugin is allowed to use, then returns an ordered list of eligible providers for each request. Each admitted attempt consumes capacity and cost even if it fails. Payment goes to the first successful provider. Capital cost is paid each epoch, including when capacity is withdrawn.

2.2 Exact settlement

For request r , let $j(r)$ be the serving provider, if any; $A(r)$ the attempted providers; P_r its payment; $C_r = \sum_{i \in A(r)} c_i$ resource cost; and T_r total latency. User utility and pre-capital welfare are

$$u_r = \begin{cases} v - P_r - \ell_L T_r, & j(r) \neq \emptyset, \\ -\ell_F - \ell_L T_r, & j(r) = \emptyset, \end{cases} \quad (1)$$

$$w_r = \begin{cases} v - C_r - \ell_L T_r, & j(r) \neq \emptyset, \\ -\ell_F - C_r - \ell_L T_r, & j(r) = \emptyset. \end{cases} \quad (2)$$

Provider profit is revenue minus attempt and capital costs.

Theorem 1 (Settlement reconciliation). *For every seed, action profile, demand realization, capacity profile, router plugin, and fallback path generated by the kernel,*

$$W_t = U_t + \sum_i \pi_{it}. \quad (3)$$

No reliability, independence, or nonbinding-capacity assumption is required.

The identity is enforced in tests, as are capacity conservation, nonnegative payments, attempted-provider ordering, and deterministic replay. If a router fee or SLA transfer is added, it enters both sides explicitly.

2.3 Parallel RL interface and information boundary

The environment exposes

$$\text{reset}(\text{seed}) \rightarrow \{i : o_{i0}\}, \quad \text{step}(\text{actions}, \text{demand}) \rightarrow (o', r, \text{done}, \text{info}, \text{result}).$$

The observation o_{it} contains epoch, demand, public quotes, and i 's own profit, attempts, successes, technical failures, and capacity rejections. It does not contain rival technology ξ_{-i} . The complete market result remains available to an evaluator. This separation prevents a common simulation error: training agents on planner-only data and then interpreting their behavior as a market equilibrium.

The API is dependency-light and can be adapted to PettingZoo's parallel interface. It deliberately treats one epoch as a simultaneous provider move. Within-epoch routing and fallback occur after actions are committed; router state advances exactly once per epoch.

2.4 Mechanism and strategy plugins

The baseline inverse-price router uses $x_i \propto p_i^{-\eta}$. This is a counterfactual mechanism, not an empirical claim about OPENROUTER. Adaptive routers may additionally use reliability or quality. The hardened rule includes five defenses: lagged smoothed inputs, provider-specific leave-one-out parameters, multi-epoch commitments, operator-neutral exploration and share caps, and a one-sided log-share trust region. Allocation gains are capped; providers are not insured against losing share after raising price or quality risk.

Strategies share the same action interface. Static policies reproduce fixed menus; species policies reproduce descriptive price regimes; Q and UCB policies learn from own reward; and scripted adversaries search quote, capacity, outage, coalition, and identity-splitting actions. A strategy name is never treated as a claim about a live provider.

3 Property Testing and Transport

Definition 1 (Property ladder). *A simulator claim at level k may use evidence from levels $1, \dots, k$ but may not inherit claims from a higher level:*

1. *accounting and transition invariants;*
2. *information-set and intervention fidelity;*
3. *held-out empirical moments;*
4. *adversarial and heterogeneous-strategy robustness;*
5. *prospective transport to owned or partner routing outcomes.*

Level 1 is exact software validation. Level 2 asks whether an intervention changes only its declared object. Level 3 asks whether the environment preserves properties visible in held-out data. Level 4 tests whether a result survives agents unlike the focal learner and profitable deviations outside its path. Level 5 is required for a live mechanism or welfare claim.

3.1 Current market properties

The latest hosted aggregate revision spans 7–21 July 2026. In its frozen provider-regime analysis, 366 training provider-model pairs produce 358 evaluable holdout pairs. Overall regime persistence is 0.944. Anchor, premium, and static discount regimes persist at 1.000, 0.992, and 0.976; active undercutting persists in 5 of 21 pairs (0.238). The regimes describe quote paths, not cost or intent.

An active-undercutter response screen contains 1,052 paired timing events in 11 provider-model clusters. The registered response-minus-shifted-placebo difference is 0.0019 with cluster interval $[-0.0438, 0.0410]$ and one-sided exact $p = 0.468$. The current owned delegated-selection match contains 81 anchor and 92 premium selections, with no active or static selections. This is support for our workload, not market share. Zero provider-model cells have a valid capacity- utilization measure, and dumping is not identified.

An independent elasticity audit rejects the claim that the public inverse-square formula is realized allocation ($z = 10.06$). Consequently, the environment does not use public shadow shares as a live calibration target. It uses them only to define explicit counterfactual router plugins.

3.2 Intervention fidelity

The signal experiment starts from an $T \times n$ signal matrix Z . Its treatment arm preserves the original order; its control independently permutes each provider column.

Proposition 1 (Exact marginal preservation). *For every provider i and every measurable function h ,*

$$\sum_{t=1}^T h(Z_{ti}) = \sum_{t=1}^T h(\tilde{Z}_{ti}). \quad (4)$$

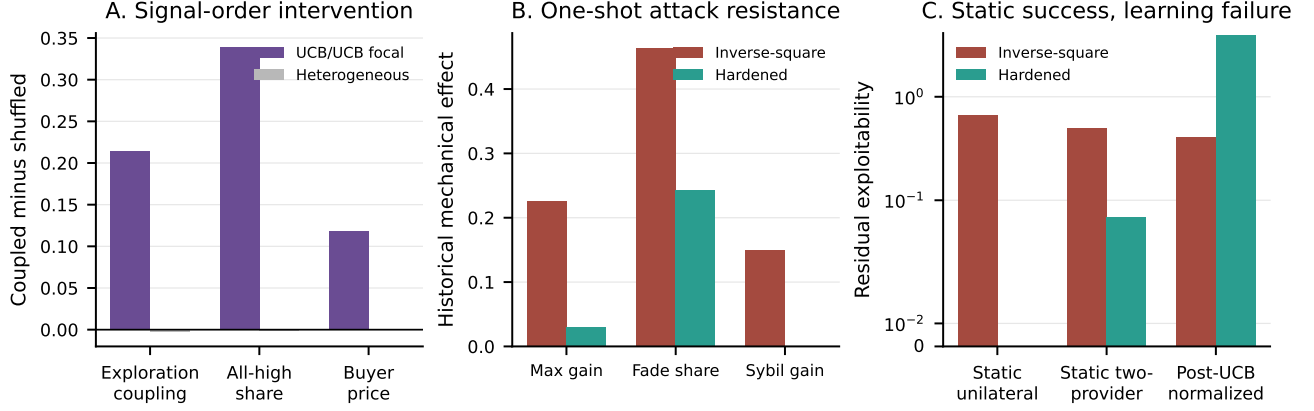


Figure 1: Property tests expose conditional success. Panel A: exact marginal-preserving signal intervention; focal UCB/UCB effects vanish across heterogeneous strategies. Panel B: the hardened adaptive router reduces mechanical one-step attacks on historical menus. Panel C: finite-grid static audits pass, but normalized post-UCB exploitability is higher. Panels B–C are simulator or menu-replay quantities, not live provider profit.

Thus the intervention preserves each finite-sample marginal multiset exactly and changes only cross-provider and temporal ordering.

Paired seeds reuse action-policy RNG and structural primitives. The intervention therefore identifies a signal-order effect inside the simulator. It does not identify the signals, beliefs, or algorithm of a real provider.

4 Experiments

4.1 E1: common signals and heterogeneous learners

Two providers choose low price 0.65 or high price 1.00 at marginal cost 0.20. With router exponent 5, the one-period payoffs satisfy the prisoner’s-dilemma ordering

$$T = 0.4032 > R = 0.4000 > P = 0.2250 > S = 0.0832.$$

The factorial varies signal-to-noise, common correlation, exponentially weighted reward memory, router exponent, learner mixture, and seed. UCB/UCB is the focal mechanism; UCB/epsilon, epsilon/epsilon, and UCB/static are transport screens. Each coupled cell is paired with its marginal-preserving shuffle.

The successful hosted run contains 1,440 factorial rows and 720 paired cells. In the focal positive-correlation, SNR-at-least-two, exponent-five subset, coupled signals raise exploration-action correlation by 0.214, joint high-price share by 0.339, and buyer price by 0.118. Buyer price increases in 69.4% of focal cells. The primary focal sign screen passes.

The heterogeneous screen fails. Its corresponding effects are -0.0030 , -0.0015 , and -0.00047 ; buyer price increases in 49.1% of cells. Across all paired cells the mean buyer-price effect is 0.0131 and the positive share is 49.9%. We therefore report a causal mechanism example and a failed universality claim. A simulator that published only the focal UCB panel would give the wrong design impression.

4.2 Empirical HMP bridge

The empirical bridge is an ordered family: (SC1) excess residual quote coupling; (SC2) preperiod routing signal-to-noise gradient; (SC3) coupled-versus-independent elasticity wedge; and (SC4) forward transition toward premium states. The latest aggregate monitor has 766 completion-price events, seven mechanism days, 19 provider-pair/model clusters, and 70 covered delegated choices.

SC1’s observed residual covariance is 0.02023, exceeding the mean of 2,000 clock-preserving circular shifts by 0.02560. Its one-sided $p = 0.04998$ is nominal, but the maximum provider-pair event share is 0.873 against a frozen cap of 0.20. SC2 and SC3 are power-gated. SC4’s descriptive premium-transition association has $p = 0.129$. The Holm family is incomplete and no claim promotes. This is not semantic caution: one more day moved SC1 across 0.05 while worsening pair concentration. A fixed property gate prevents optional causal language.

4.3 E2: static defense versus adaptive learning

The adaptive-router screen uses 4,878 eligible historical menus, 652,680 historical quote-attack rows, 200 strategic menus, 7,200 cost-by-capacity-by-rule cells, 32 sequential-response cells, 320 UCB rows, and 40 bounded Q-learning runs. All artifacts share one immutable input revision.

Against inverse-square routing, hardening reduces mean maximum allocation gain from 0.2262 to 0.0300, share captured by a 25% fading quote from 0.4638 to 0.2422, and two-identity combined-share gain from 0.1499 to zero. Model-day cluster intervals exclude zero in the favorable direction. Finite-grid mean unilateral exploitability falls from 0.6664 to zero; mean two-provider exploitability from 0.4976 to 0.0690. All eight hardened sequential best-response cells converge with zero residual gain within the declared grid.

The learning screen reverses the normalized ranking. Post-UCB residual exploitability is 3.9291 under hardening and 0.4057 under inverse-square, a 9.68 ratio. Absolute gains are much closer (5.89×10^{-4} versus 3.64×10^{-4}), so lower provider profit contributes to the normalized failure. The original UCB seed rows were deterministic duplicates; the correct independent unit is eight menus. A later stochastic engineering audit restores distinct seed paths and finds lower hardened absolute deviation gain on seven of eight menus, but still fails the frozen normalized gate. Neither result is confirmatory.

The design lesson is concrete. A one-sided trust region can prevent a provider from suddenly gaining traffic by undercutting, yet introduce committed state that learners estimate slowly. Static strategy-proofness on a grid does not imply low learning regret or fast convergence in a nonstationary repeated game.

5 Mechanism Evaluation Without Welfare Theater

The kernel computes user utility, provider profit, and social surplus for every declared scenario. Those numbers are internally exact and externally conditional on workload value, provider cost, and capacity. Current public data identify none of these primitives well enough for a market-wide welfare estimate.

We therefore recommend a robust design problem. Let θ parameterize score weights, commitments, exploration, caps, and fees; Ω be the set of cost, quality, capacity, and learning models consistent with observed properties. The evaluator solves

$$\min_{\theta} \sup_{\omega \in \Omega} R_U(\theta, \omega) + \lambda_1 E_1(\theta, \omega) + \lambda_2 E_2(\theta, \omega) + \lambda_V V(\theta, \omega), \quad (5)$$

subject to completion, concentration, budget, and delivered-capacity floors. R_U is user regret; E_1 and E_2 are unilateral and two-provider exploitability; V is constraint violation. Results should be reported as a Pareto frontier, with cost/value sensitivity, rather than a single welfare point.

The current replay candidate uses price exponent 1.25 and exploration 0.10. On 18,947 held-out hourly menus it lowers a quote-coupling proxy by 49.3%, raises displayed price by 2.08%, and changes public reliability by 0.0059 percentage points. Provider behavior is fixed. The result is sufficient to choose a prospective randomized arm, not to deploy the rule or claim welfare.

The next experiment randomizes production delegation, price-ordered fallback, and the frozen candidate score within model-prompt-time blocks. Primary outcomes are attempted-request spend, completion, end-to-end latency, and blinded task fidelity. Capacity commitments require a separate provider-facing trial. A scalar welfare statement is promoted only over a preregistered box of user values and provider costs.

6 Related Work

General multi-agent interfaces and game-learning suites prioritize broad domain coverage [Terry et al., 2021, Lanctot et al., 2019]. Our contribution is a market-specific settlement layer and an empirical transport standard. Algorithmic-pricing experiments show that independent learners can sustain high prices [Calvano et al., 2020]. Hansen, Misra, and Pai identify a distinct channel through correlated price experiments [Hansen et al., 2021]; our exact signal-order intervention isolates that channel and shows its learner dependence.

Brown and MacKay study heterogeneous pricing algorithms and repricing frequency [Brown and MacKay, 2023]. Our persistent quote regimes are compatible with that motivation but do not identify algorithms. Johnson, Rhodes, and Wildenbeest show how platform prominence changes algorithmic price competition [Johnson et al., 2023]; our adaptive-rule experiments add hidden capacity, fallback, and learning-speed failures. Model-routing benchmarks choose among models [Chen et al., 2023, Ong et al., 2024, Hu et al., 2024]; we hold the model label fixed and study competition among execution providers.

7 Limitations, Ethics, and Conclusion

The public panel is measured in weeks. Owned probes cover one account and workload distribution. Public enforcement is aggregated. Provider cost, market-wide demand, user value, private rebates, and deliverable capacity are unobserved. UCB, epsilon-greedy, Q-learning, and static policies span only a small subset of strategic behavior. Grid exploitability is not unrestricted dynamic exploitability. The hardened-rule screen is mixed and not confirmatory.

Paid probes are assignment-first, minimal-token, budget-capped, and remotely rate-limited. Public artifacts remove request-level rows. The research does not send adversarial load, infer prompt contents, or identify provider employees. Provider labels are descriptive regimes, not allegations. We do not claim collusion, communication, dumping, or literal front-running.

The central finding is that routing-market simulation needs a refusal mechanism. The focal UCB result is real inside its game and fails across heterogeneous agents. The hardened router resists one-step manipulation and fails its first learning gate. The public quote signal is nominal and fails concentration and family completion. A property ladder makes those boundaries executable. That is what turns a strategic simulator from a storytelling device into a scientific instrument.

A Proofs

Proof of theorem 1. For a successful request, user utility plus the serving and attempted providers' variable profit is $(v - P_r - \ell_L T_r) + (P_r - C_r) = v - C_r - \ell_L T_r = w_r$. For a failed request it is $(-\ell_F - \ell_L T_r) - C_r = w_r$. Sum over requests and subtract each provider's epoch capital cost on both sides. $\square$

Proof of proposition 1. For each i , $\tilde{Z}_{\cdot i}$ is a permutation of $Z_{\cdot i}$. Summation is invariant to permutation. $\square$

B Reproducibility and Evidence Ledger

The RL wrapper is `src/orcap/market_env/rl_env.py`; its tests verify private-information boundaries, reward reconciliation, horizons, invalid calls, and exact reset replay. A documented four-epoch example at `examples/strategic_routing_demo.py` is itself tested for byte-equivalent seeded output and settlement reconciliation; the adjacent environment guide specifies the observation contract and custom-router path. Kernel tests additionally cover capacity, failure, fallback, transfers, and stable random substreams. Signal tests verify exact column multisets and paired interventions. Adversarial tests cover quote monotonicity, share caps, commitment state, identity grouping, unilateral grids, and coalition enumeration.

The WF18 hosted evidence is GitHub Actions run 29834114219, code revision 4e89018, and dataset revision e4174d08. The adaptive screen is run 29780088479, code revision b7908e09, and dataset revision 0a7c6bef. Aggregate manuscript inputs are recorded in `papers/neurips/hosted-evidence.json`; the figure is generated by the adjacent build script. The hosted WF18 job hydrates one immutable snapshot, runs WF16 and WF18, executes the two-seed simulator, validates privacy boundaries, removes owned request rows, publishes aggregates to Hugging Face, refreshes a private dashboard, and preserves an immutable artifact.

C Environment Card

Agents and actions. The agents are inference providers. A simultaneous action is a positive quote, an admitted-capacity fraction in $[0, 1]$, and an availability state. The core environment does not let a provider change its private marginal cost, physical capacity, reliability, or base latency during an episode. Experiments that model investment do so in an outer stage and create a new immutable provider specification.

Observations and rewards. Every provider sees the epoch, realized demand, the public quote vector, and its own preceding profit, attempts, completions, failures, and capacity rejections. The default reward is own profit after variable and installed-capacity cost. Alternative rewards must be named by an experiment; the environment never hands social welfare to a provider by default. The evaluator sees request outcomes, user utility, provider profit, resource cost, payment, and conditional welfare.

Transition and termination. Within an epoch, the router orders candidates, capacity clips attempts, reliability draws determine technical failure, and fallback continues to the declared limit. The episode ends at a fixed horizon; there is no outcome-based early stopping. Calling `step` after termination is an error. Resetting with the same seed exactly replays the same transition under identical actions.

Randomness and pairing. A stable hash of base seed, epoch, request, event type, and provider creates disjoint random substreams. Mechanism comparisons reuse the same base seed so a routing change does not silently redraw reliability. Signal interventions have a separate, recorded seed and assert column-multiset equality before running.

Known failure modes. Expected-allocation training removes routing noise and can create duplicate seed paths. Normalized exploitability is unstable near zero provider profit. A finite quote grid can miss continuous or history-dependent deviations. Public menus cannot calibrate hidden eligibility or market-wide demand. Static action robustness does not imply learning convergence. The released screen reports all five failures rather than treating them as tuning opportunities.

Minimum reporting standard. A routing-market experiment should publish: code and scenario hashes; action and observation schemas; provider information boundaries; seed and pairing policy; router state-transition timing; transfer reconciliation; capacity and fallback semantics; calibration/holdout split; algorithm families; global and coalition deviation class; absolute and normalized exploitability; failed transport gates; and a statement of which welfare primitives are assumed rather than measured.

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